
Data Structures

BS (CS) _Fall_2024

Lab_07 Tasks



Learning Objectives:

1. Doubly Linked List
2. Circular Linked List

Lab Tasks

Submission Instructions

1. Create a separate **.h** and **.cpp** file for each class.
2. Now create a new folder with name *ROLLNO_SEC_LAB07* e.g. **i23XXXX_A_LAB07**(Also name your **cpp** file like this.)
3. Move all of your files to this newly created directory and compress it into a **.zip file**. Return the test file as well.
4. Now you have to submit this zipped file on Google Classroom.
5. If you don't follow the above-mentioned submission instruction, you will be marked **zero**.
6. Plagiarism or AI generated code in the Lab Task will result in **zero** marks in the whole category.

Question 1

The Josephus problem is a famous theoretical problem. N people, numbered from 1 to N , are seated in a circle, represented by a **doubly circular linked list**. Starting from the first person, a hot potato is passed around. After M passes, the person holding the hot potato is eliminated from the circle. The circle then closes ranks, and the game continues from the next person (the one sitting immediately after the eliminated person). The process is repeated until only one person remains, who is declared the winner.

For example:

- If $M = 0$ and $N = 5$, players are eliminated in sequential order, and player 5 wins.
- If $M = 1$ and $N = 5$, the elimination order is 2, 4, 1, 5, and player 3 wins.

Task:

Write a program using a **doubly circular linked list** to simulate the Josephus problem for arbitrary values of M and N . Ensure that your program handles the deletion of nodes (representing people) efficiently, and that the doubly circular structure is properly maintained throughout the process.

M	N	Output
2	6	1
1	5	3
3	9	1